
FedSLIP: Federated Sparse LoRA Identity Personalization for Heterogeneous Language Models

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Code Repository: <https://github.com/TayyabWaseem/FedSLIP.git>

Abstract

Federated learning (FL) enables language models to adapt from decentralized private data, but severe client heterogeneity makes a single shared adapter insufficient for personalization. Standard FedAvg with Low-Rank Adaptation (LoRA) is communication-efficient, yet its single low-rank subspace must simultaneously encode global consensus and client-specific identity. This creates a collaboration penalty: global averaging dilutes local knowledge, forcing clients to perform additional local fine-tuning after federation to recover personalized performance. We propose **FedSLIP**, Federated Sparse LoRA Identity Personalization, a dual-track PEFT method that prevents client identity from slipping away during aggregation. Each adapted Transformer projection contains a rank-8 global LoRA path aggregated by FedAvg with FedProx regularization and a rank-32 private LoRA path that remains strictly on-device. The private path is sparsified by dynamic Lottery Ticket Hypothesis magnitude masking at 75% sparsity, giving each client a compact personalized identity subnetwork with zero extra communication. FedSLIP further stabilizes training through separate AdamW optimizers, explicit gradient detachment during private updates, a 10-round warmup for mask discovery, and communication filtering that transmits only global adapters. On non-IID AG News with four 80% label-skewed clients, FedSLIP improves zero-shot personalized accuracy from 90.43% for FedAvg+LoRA to 93.89%, nearly matching FedAvg+LoRA after extra K-step personalization (94.24%), while retaining the same 4.50 MB per-round communication cost as vanilla LoRA.

1. Introduction

Federated learning (FL) is a distributed training paradigm in which clients collaboratively optimize a model without transferring raw local data to a central server (1). This is essential for privacy-sensitive language applications, where user text may contain medical, financial, legal, or behavioral information. However, applying FL to modern pre-trained language models (PLMs) is difficult. Full fine-tuning requires updating and communicating millions to billions of parameters, which is not practical for cross-device or resource-constrained settings.

Parameter-efficient fine-tuning (PEFT) addresses this bottleneck by freezing the pretrained backbone and training only a small set of adaptation parameters. Low-Rank Adaptation (LoRA) is one of the most widely used PEFT methods: it injects trainable low-rank matrices into selected linear layers while leaving the original model weights fixed (2). In federated settings, LoRA allows clients to transmit adapter parameters rather than full model weights. This makes FL over language models feasible, and recent federated PEFT work has validated this direction for NLP and instruction-tuning tasks.

Feasibility, however, does not solve personalization. Real federated data are non-IID: each user or institution has distinct labels, styles, domains, and task preferences. When a single LoRA adapter is globally averaged, it must represent both the global consensus and each client’s local identity. These objectives conflict under severe heterogeneity. The adapter becomes a “one-size-fits-none” model: strong enough to perform globally, but unable to preserve the specialized patterns required by individual clients. In our baseline implementation on AG News, FedAvg+LoRA obtains 90.06% global accuracy and 90.43% average client accuracy, but clients require an extra K-step local personalization phase to reach 94.24%.

This project developed in three stages. First, we implemented a standard FedAvg+LoRA baseline using DistilBERT, rank-8 LoRA adapters on query and value projec-

tions, four AG News clients, and 80% dominant-label skew per client. This established the collaboration penalty: federation improved global sharing but diluted local identity. Second, we introduced an initial dual-adapter design with a global path and a private path. This validated the idea that global knowledge and local identity should be separated, but it also exposed two issues: optimization paths remained entangled through shared gradients, and communication accounting could be harmed if non-adapter parameters were serialized. Third, we implemented the final FedSLIP method, which cleanly decouples architecture, optimization, sparsity, and communication.

FedSLIP is based on a simple design principle: *what is shared should be low-rank, stable, and aggregated; what is personal should be expressive, sparse, and private*. The global track is a rank-8 LoRA adapter trained with FedAvg and FedProx. The private track is a rank-32 LoRA adapter, never transmitted, sparsified by dynamic magnitude masks inspired by the Lottery Ticket Hypothesis (LTH). During inference, each client uses both tracks, giving zero-shot personalization without post-training local fine-tuning.

Our contributions are:

- We introduce FedSLIP, a dual-track federated LoRA architecture that separates global consensus from client-private sparse identity inside Transformer attention projections.
- We design a rank-32 private LoRA path with 75% dynamic LTH magnitude masking, producing a compact client-specific identity subnetwork that remains local.
- We stabilize heterogeneous training with separate global and private AdamW optimizers, explicit gradient detachment, FedProx regularization with $\mu = 0.1$, a 10-round warmup phase, and private loss weighting $\lambda_p = 0.6$.
- We enforce strict communication filtering so only rank-8 global LoRA adapters are uploaded and downloaded; private adapters and classifier heads remain local.
- We empirically show that FedSLIP achieves 93.89% zero-shot personalized accuracy on non-IID AG News, improving over FedAvg+LoRA zero-shot personalization while preserving the same 4.50 MB per-round communication budget.

2. Related Work

Federated optimization under non-IID data. FedAvg established that deep models can be trained from decentralized data through local training followed by server averaging (1). Its efficiency comes from performing multiple local

updates per communication round, but this also increases client drift when local objectives differ. FedProx adds a proximal term to constrain local models around the server model, improving robustness under statistical and systems heterogeneity (3). SCAFFOLD uses control variates to correct drift more explicitly (4). These methods identify the same central challenge: non-IID clients can pull the model in inconsistent directions. FedSLIP adopts this insight but applies drift control only to the aggregated global LoRA path, because the private path is intentionally allowed to specialize.

Federated foundation models and PEFT. The rise of foundation models has shifted FL research toward communication-efficient adaptation. Federated instruction tuning shows that decentralized user data can improve language models without centralizing private examples (8). However, full-parameter adaptation is expensive, so PEFT methods are increasingly central. LoRA reduces trainable parameters by representing updates as low-rank decompositions (2). SLoRA evaluates federated PEFT for language models and highlights that sparse or low-density adaptation can approach full tuning while reducing training cost (9). pFedLoRA studies personalization and model heterogeneity, showing that LoRA can support clients with different architectures or resource constraints (10). FedSLIP complements these methods by targeting a specific gap: the lack of strict separation between globally aggregated adapter knowledge and private client identity.

Personalized federated learning. Personalized FL recognizes that a single global model is often suboptimal for every client. Per-FedAvg learns a global initialization that can be rapidly adapted by each client (5). APFL interpolates between global and local models (6). Ditto trains personalized models regularized toward a robust global model (7). These methods show that personalization and global learning are compatible. However, many personalized FL methods assume full-model training or additional local adaptation after federation. FedSLIP implements personalization within PEFT itself: the global adapter is shared, while the private adapter continuously learns local residual identity during federation.

Comparison to FedDLP. FedDLP-style federated deep learning personalization also motivates the use of client-specific structure, but it is conceptually closer to full-model or large-module personalization than to PEFT for frozen language models. Its pruning mechanism is primarily an efficiency device for selecting important components of a personalized model. In contrast, FedSLIP operates inside DistilBERT’s LoRA-adapted attention projections, keeps the pretrained backbone frozen, and treats sparsity as an identity-forming mechanism: the private sparse adapter is

not a compressed upload but a client-local residual subnetwork that is never aggregated. This difference is important for federated language models because it preserves the communication profile of LoRA while still allowing each client to retain a private task-specific representation.

Sparse subnetworks and lottery tickets. The Lottery Ticket Hypothesis states that dense networks contain sparse subnetworks that can match the performance of the full model when trained appropriately (11). Magnitude pruning is a classical method for identifying important weights (12). In FL, pruning has often been used to reduce communication (13). FedSLIP uses sparsity differently. Since the private adapter is never transmitted, the purpose of sparsity is not primarily payload reduction, but regularized identity formation. The sparse mask forces each client to identify a compact task-specific subnetwork within a higher-rank private adapter.

3. Problem Formulation

We consider a federated classification setting with K clients. Client k owns local dataset

$$\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}, \quad (1)$$

sampled from private distribution $P_k(x, y)$. The global dataset is $\mathcal{D} = \cup_{k=1}^K \mathcal{D}_k$ with $n = \sum_k n_k$. Under non-IID label skew, $P_k(y)$ differs across clients. In our setup, each client has one dominant AG News class with ratio $\rho = 0.8$.

The standard FL objective is

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{(x,y) \in \mathcal{D}_k} \ell(f_{\theta}(x), y) \quad (2)$$

where ℓ is cross-entropy loss. This objective learns a single parameter vector θ for all clients. Under severe heterogeneity, such a model may perform well on a balanced global test set while failing to capture local client preferences.

FedSLIP decomposes trainable parameters into a globally shared adapter θ_g and client-private adapter $\theta_{p,k}$. The pre-trained backbone θ_0 is frozen. Client k predicts with

$$f_k(x) = f(x; \theta_0, \theta_g, \theta_{p,k}). \quad (3)$$

The server aggregates only θ_g :

$$\theta_g^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{g,k}^{t+1}. \quad (4)$$

Each private adapter $\theta_{p,k}$ remains local for all rounds. The

personalized objective can be written as

$$\min_{\theta_g, \{\theta_{p,k}\}} \sum_{k=1}^K \frac{n_k}{n} F_k(\theta_0, \theta_g, \theta_{p,k}) \quad \text{s.t.} \quad \theta_g \text{ is shared, } \theta_{p,k} \text{ is private and sparse} \quad (5)$$

The main challenge is preventing the private path and global path from co-adapting destructively during local optimization.

4. Methodology

4.1. Baseline: FedAvg with LoRA

For a frozen linear layer $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, standard LoRA adds a low-rank update:

$$h = W_0 x + \frac{\alpha}{r} B A x, \quad (6)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are trainable. The baseline uses rank $r = 8$, scaling $\alpha = 16$, dropout 0.1, and adapts DistilBERT’s `q_lin` and `v_lin` modules. Clients train local LoRA adapters for two local epochs, upload adapter weights, and the server averages them.

This baseline is strong globally but limited personally. Since the same adapter carries both global and local information, averaging suppresses client-specific directions. The README identifies this as the collaboration penalty: clients typically require additional K-step local fine-tuning after federation to recover personalized performance.

4.2. Design Evolution and Key Innovations

FedSLIP was developed through three implementation stages, each isolating a different failure mode of federated PEFT under label skew. The baseline stage used a single rank-8 LoRA adapter with FedAvg. It provided a strong global reference point, but its client accuracy revealed that collaborative averaging alone did not preserve local identity. The first improvement introduced an additive global-private adapter structure, moving from partial sharing to an explicit decomposition of knowledge. This improved the conceptual personalization mechanism, but the implementation still optimized both paths through one training graph and incurred a larger communication footprint. The final FedSLIP implementation keeps the dual-track idea but makes the separation operational: the global path is optimized and transmitted independently, while the private path is optimized as a detached sparse residual that never leaves the client.

4.3. FedSLIP DualLoRALinear

FedSLIP replaces each target projection with a `DualLoRALinear` module. It contains:

Table 1. Design progression from the notebooks. “FT” denotes post-training local fine-tuning.

Stage	Main change	Comm.
Baseline	Single global LoRA trained by FedAvg; K-step FT required for strongest personalization.	4.50 MB
First Improvement	Added private LTH LoRA, but paths remained jointly optimized and communication boundary was loose.	11.31 MB
FedSLIP	Separate global/private optimizers, FedProx on global path, detached private update, warmup masking, and global-only transmission.	4.50 MB

- a frozen base layer W_0 ,
- a rank-8 global LoRA path (A_g, B_g) ,
- a rank-32 private LoRA path (A_p, B_p) ,
- binary private masks (M_A, M_B) .

For input x , the module computes

$$z_0 = W_0 x, \quad (7)$$

$$z_g = \frac{\alpha_g}{r_g} B_g A_g x, \quad (8)$$

$$z_p = \frac{\alpha_p}{r_p} (M_B \odot B_p) (M_A \odot A_p) x. \quad (9)$$

The personalized forward pass is

$$h_k = z_0 + z_g + z_p. \quad (10)$$

For global-only training or global evaluation, the private path is bypassed:

$$h_g = z_0 + z_g. \quad (11)$$

For private adaptation, the global residual is detached:

$$h_p = z_0 + \text{stopgrad}(z_g) + z_p. \quad (12)$$

This detachment is central. The private adapter can learn residual client identity using the global representation as context, but its gradients cannot modify the global adapter.

4.4. Sparse Private Identity with LTH Masking

The private adapter uses higher rank ($r_p = 32$) than the global adapter ($r_g = 8$) so each client has enough local capacity. To avoid uncontrolled memorization, FedSLIP applies 75% magnitude sparsity. For private weight tensor W , the mask is

$$M_{ij} = \mathbb{I}\{|W_{ij}| \geq \tau_s(W)\}, \quad (13)$$

where $\tau_s(W)$ is the s -percentile absolute-value threshold and $s = 0.75$. Thus, only the top 25% magnitude weights remain active.

This design creates a dense-to-sparse identity search. The rank-32 private adapter provides a wide search space; the LTH mask selects the strongest local subnetwork. During the first $T_w = 10$ rounds, masks are dynamically updated to allow useful magnitudes to emerge. After warmup, the sparse private pathway serves as the client’s zero-shot personalized identity.

4.5. Local Optimization

Each client performs two sequential optimization steps per batch.

Global consensus update. The client disables the private path and optimizes only global LoRA parameters. FedProx regularizes the local global adapter around the server adapter:

$$\mathcal{L}_{g,k} = \ell(f(x; \theta_0, \theta_{g,k}), y) + \frac{\mu}{2} \|\theta_{g,k} - \theta_g^t\|_2^2, \quad (14)$$

with $\mu = 0.1$. This reduces client drift in the only parameters that will be aggregated.

FedProx is applied only to the global LoRA path because this is the only state that participates in server-side averaging. The proximal term therefore constrains the parameters whose drift would directly corrupt the shared model. Applying the same penalty to the private path would work against the purpose of personalization: the private adapter must be free to move toward the client’s local label skew, while the global adapter must remain close enough to the server reference to support stable aggregation across heterogeneous clients.

Private identity update. The client then enables the private path and detaches the global residual:

$$\mathcal{L}_{p,k} = \lambda_p \ell(f(x; \theta_0, \text{stopgrad}(\theta_{g,k}), \theta_{p,k}), y), \quad (15)$$

where $\lambda_p = 0.6$. Private parameters are optimized using a separate AdamW optimizer with half the global learning rate. This prevents private learning from overwhelming shared representation learning.

4.6. Communication Filtering

FedSLIP strictly transmits only parameters whose names match the global adapter matrices: `lora_A_global` and `lora_B_global`. Private adapters, private masks, classifier heads, and pre-classifier heads remain local. For adapted layers $\mathcal{S}$, the per-client payload is

$$C_{\text{client}} = 4 \sum_{\ell \in \mathcal{S}} r_g (d_{\text{in}}^\ell + d_{\text{out}}^\ell) \text{ bytes}, \quad (16)$$

Algorithm 1 FedSLIP local training at client k

```

1: Receive global LoRA  $\theta_g^t$  from server
2: Synchronize only  $(A_g, B_g)$ ; keep  $\theta_{p,k}$  local
3: for local epoch  $e = 1, \dots, E$  do
4:   for mini-batch  $(x, y)$  do
5:     Disable private path and compute  $\mathcal{L}_{g,k}$ 
6:     Update  $\theta_{g,k}$  using global AdamW and FedProx
7:     Enable private path with detached global residual
8:     Update  $\theta_{p,k}$  using private AdamW and  $\lambda_p \mathcal{L}_{p,k}$ 
9:   end for
10: end for
11: if  $t \leq T_w$  then
12:   Update private masks by magnitude thresholding
13: end if
14: Upload only global LoRA matrices  $(A_g, B_g)$ 
    
```

assuming float32 communication. The total per-round cost includes upload and download across clients:

$$C_{\text{round}} = 2KC_{\text{client}}. \quad (17)$$

Because $\theta_{p,k}$ is never transmitted, FedSLIP adds zero communication relative to the global LoRA channel.

5. Experiments and Results

5.1. Experimental Setup

We evaluate on AG News, a four-class text classification dataset containing World, Sports, Business, and Sci/Tech categories. We simulate four clients, each associated with one dominant class. Each client receives 2,000 training examples and 500 test examples with dominant ratio $\rho = 0.8$. The balanced global test set uses 400 examples per class. The model is `distilbert-base-uncased`. Training uses 30 communication rounds, two local epochs per round, batch size 32, learning rate 10^{-4} , weight decay 0.01, and gradient clipping at norm 1.0.

Dataset and partitioning. AG News is used because its four labels naturally support a controlled client-identity experiment: each client can be assigned one dominant topic while retaining a small amount of data from the remaining topics. For client k , 80% of the local train and test split is sampled from the client’s dominant class and the remaining 20% is sampled from the other three classes. This partitioning creates a severe but interpretable non-IID setting in which personalization can be measured directly by evaluating each client on its own skewed test distribution. The global test loader is balanced across all four classes, so global accuracy measures shared generalization rather than dominance of any one client.

Model architecture. All methods use DistilBERT for sequence classification with the pretrained backbone frozen except for adapter and task parameters required by each method. LoRA modules are inserted into the self-attention query and value projections (`q_lin` and `v_lin`), matching the baseline notebook and the final FedSLIP implementation. FedSLIP keeps the same rank-8 global LoRA path as the baseline and adds a client-local rank-32 private path with masked weights. This isolates the impact of the personalization mechanism from changes in the pretrained backbone.

We compare:

- **Local-Only:** each client trains independently without federation.
- **FedAvg+LoRA:** standard rank-8 LoRA with FedAvg.
- **First Improvement:** initial dual-track design.
- **FedSLIP:** final sparse dual-track method with clean decoupling.

Evaluation metrics. We report five metrics. Global balanced accuracy evaluates the server-visible global path on a balanced AG News test set. Per-client global accuracy evaluates the global path on each client’s skewed test set and exposes which clients are underserved by aggregation. Zero-shot personalized accuracy evaluates each persistent FedSLIP client model with both global and private paths active, without any additional post-training fine-tuning. K-step personalized accuracy is reported for the FedAvg+LoRA baseline after local adaptation and serves as a strong personalization reference. Finally, communication cost is measured as upload plus download bytes per round for the transmitted trainable parameters.

5.2. Global Accuracy

Table 3 reports balanced global test accuracy. FedAvg+LoRA achieves the highest global score because all trainable capacity is optimized for shared performance. FedSLIP is slightly lower globally, but it remains stable and substantially improves over the first dual-track attempt. This validates the final design: private capacity can be added without collapsing the global path, provided the optimization and communication boundaries are clean.

The first improvement is especially informative. It shows that simply adding a private adapter is not sufficient. Without strict gradient detachment and optimization separation, the global path can absorb unstable interactions from client-specific learning. FedSLIP restores stable global behavior through FedProx, global-only updates, and detached private adaptation.

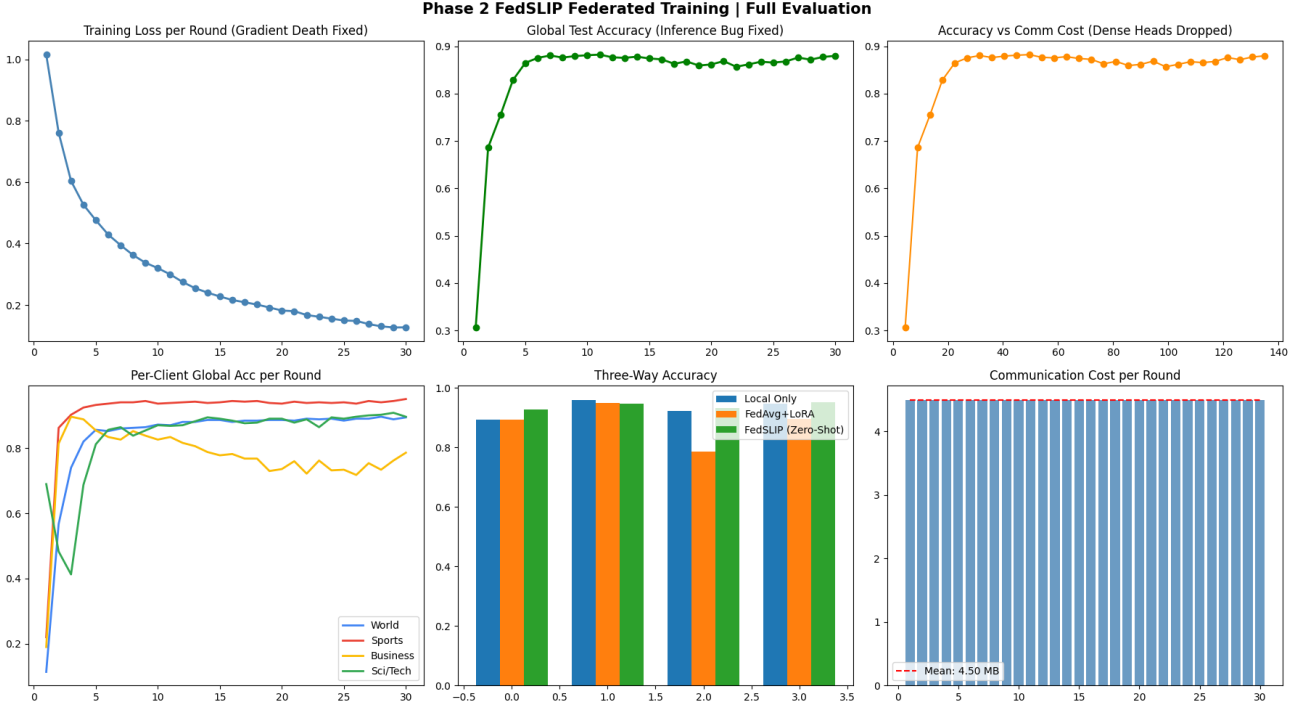


Figure 1. Phase 2 FedSLIP full evaluation over 30 federated rounds. The figure summarizes training loss, global balanced accuracy, accuracy versus cumulative communication cost, per-client global accuracy, three-way personalization accuracy, and per-round communication cost. FedSLIP converges stably, preserves strong zero-shot personalization, and maintains a constant 4.50 MB per-round communication budget after dropping dense classifier heads from transmission.

5.3. Zero-Shot Personalization

The primary objective is zero-shot personalization: personalized client accuracy without any additional local fine-tuning after federation. Table 4 reports per-client results.

FedSLIP improves average zero-shot personalized accuracy from 90.43% to 93.89%, a gain of 3.46 percentage points over FedAvg+LoRA zero-shot inference. The improvement is largest on clients most harmed by global averaging. Business improves from 87.98% to 93.19%, and Sci/Tech improves from 87.78% to 95.19%. FedAvg+LoRA reaches 94.24% only after an extra K-step local fine-tuning phase. FedSLIP nearly matches that score directly at inference time, demonstrating that the sparse private identity path successfully absorbs client-specific structure during federation.

5.4. Communication Efficiency

Table 5 reports communication cost. FedSLIP transmits 147,456 global LoRA parameters per client, matching the intended rank-8 LoRA payload. Across four clients, upload plus download costs 4.50 MB per round and 135.00 MB over 30 rounds. The private rank-32 adapter is never communicated.

The first improvement increased cost because the commu-

nication boundary was not yet clean. FedSLIP corrects this by transmitting only global adapters. This is important for practical PEFT systems, where classifier heads or other dense task-specific components can unintentionally enter the serialized payload if not filtered explicitly.

5.5. Measured Design Ablation

Table 6 summarizes the measured design progression using the values reported by the notebooks. The important trend is not that every intermediate variant improves every metric, but that each stage reveals a different bottleneck. FedAvg+LoRA is globally strong but requires K-step fine-tuning to approach the local-only personalization ceiling. The first improvement shows that private identity capacity is useful, but also demonstrates that a naive dual adapter can destabilize global learning and increase communication. FedSLIP preserves the communication budget of the baseline while recovering most of the K-step personalization benefit in a zero-shot form.

5.6. Convergence Behavior

The training trends show three distinct regimes. FedAvg+LoRA quickly reaches high global accuracy but leaves client identity underrepresented. The first improve-

Table 2. Main hyperparameters used in the final FedSLIP notebook.

Hyperparameter	Value
Backbone	DistilBERT-base-uncased
Clients	4
Dominant class ratio	0.80
Train / test samples per client	2000 / 500
Global test samples per class	400
Target modules	<code>q_lin, v_lin</code>
Global LoRA rank / alpha	8 / 16
Private LoRA rank / alpha	32 / 64
Private sparsity ratio	0.75
Warmup rounds	10
FedProx μ	0.1
Private loss weight λ_p	0.6
Rounds / local epochs	30 / 2
Batch size / learning rate	32 / 10^{-4}
Weight decay	0.01

Table 3. Global balanced test accuracy after 30 rounds.

Method	Global Accuracy
FedAvg+LoRA	90.06
First Improvement	74.25
FedSLIP Global Path	88.00

ment initially improves personalization but destabilizes global performance, with global accuracy dropping to approximately 74%. FedSLIP maintains a stable global path while allowing private adapters to specialize. Its loss decreases gradually rather than collapsing to zero, consistent with the intended behavior: the private path learns residual local structure without overwriting global semantics.

6. Discussion and Ablation

Decoupling is necessary. The difference between the first improvement and FedSLIP is the strongest ablation. Both use a dual-track idea, but only FedSLIP enforces clean separation through global-only updates, private-only updates, and detached global residuals. The first improvement reaches 92.28% personalized accuracy but suffers 11.31 MB per round and poor global accuracy. FedSLIP reaches 93.89% zero-shot personalization with 4.50 MB per round and 88.00% global accuracy.

Causal role of the design choices. The three main FedSLIP choices have complementary causal effects. Dual-track separation prevents one low-rank adapter from being forced to encode both shared semantics and client skew, which reduces destructive averaging and yields a more sta-

ble global path. Private sparsity then restricts the higher-rank local adapter to a compact subset of high-magnitude directions, which converts extra private capacity into a client identity rather than unstructured memorization. Warmup delays hard mask selection until private weights have received enough gradient signal, so the selected sparse subnetwork reflects learned local structure rather than initialization noise. Together these effects explain the observed pattern: FedSLIP sacrifices only a small amount of global accuracy relative to FedAvg+LoRA while recovering strong zero-shot per-client personalization.

Gradient detachment prevents co-adaptation. If the private loss backpropagates into the global path, the two paths can learn compensating errors. Detachment makes the global path a fixed semantic context during private learning. The private adapter must therefore learn client-specific residuals rather than reshaping shared knowledge.

Separate optimizer requirement. A single optimizer over both LoRA tracks is problematic because the global and private matrices sit inside the same DistilBERT attention projections, specifically the `q_lin` and `v_lin` modules. When one optimizer receives gradients from the combined forward pass, updates to the global and private paths are coupled through the same classification loss. Under non-IID label skew, the private path can start fitting a client-specific direction while the global path simultaneously learns a compensating direction that improves the local batch but harms the aggregated representation. Separate optimizers make the intended roles explicit: the global optimizer updates only the server-visible consensus adapter, while the private optimizer updates only the local residual identity adapter. This prevents shared-gradient co-adaptation and was a key difference between the unstable first improvement and the final FedSLIP training loop.

Diagnostic evidence. The notebooks exposed path mixing through the behavior of the initial dual-adapter implementation. When the private path was active without detached optimization, personalized accuracy improved relative to the global-only view, but the global model became unstable: global accuracy dropped to 74.25% and the per-round communication accounting rose to 11.31 MB because the implementation did not yet cleanly separate trainable and transmitted states. In Phase 2, the diagnostic checks were made explicit by evaluating the global model with the private path bypassed, grouping global and private parameters separately, and transmitting only `lora_A_global` and `lora_B_global`. The resulting recovery to 88.00% global accuracy with 4.50 MB per round supports the interpretation that the earlier failure was caused by mixed optimization and communication boundaries rather than by the dual-track idea itself.

Table 4. Per-client accuracy on non-IID local test sets. FedSLIP personalization is zero-shot: no post-training K-step local fine-tuning is performed.

Client	Dominant Class	Local-Only	FedAvg+LoRA Zero-Shot	FedAvg+LoRA + K-Step FT	FedSLIP Zero-Shot
C1	World	89.38	90.98	93.79	92.59
C2	Sports	95.99	94.99	96.19	94.59
C3	Business	92.18	87.98	92.99	93.19
C4	Sci/Tech	94.59	87.78	93.99	95.19
Average	–	93.04	90.43	94.24	93.89

Table 5. Communication cost over 30 rounds.

Method	Sent Params	MB/Round	Total MB
FedAvg+LoRA	147,456	4.50	135.00
First Improvement	294,912	11.31	339.26
FedSLIP	147,456	4.50	135.00

FedProx belongs on the global path. FedProx is applied only to global LoRA parameters because only those parameters are aggregated. Applying it to private parameters would suppress personalization. Omitting it from global parameters would expose the shared adapter to severe label-skew drift. The value $\mu = 0.1$ provides a practical stabilizing force without freezing the global path.

Higher private rank plus sparsity is effective. A rank-8 private adapter would be too constrained to model diverse local patterns. A dense rank-32 adapter could overfit. FedSLIP combines rank 32 with 75% sparsity, giving clients a large search space but forcing the final identity to be compact. This explains why the method performs well in zero-shot personalization while avoiding extra communication.

Warmup improves mask quality. The 10-round warmup allows private weights to develop meaningful magnitudes before pruning. Immediate pruning would select masks from near-initial random weights. Delayed or absent pruning would weaken the LTH-style regularization. Warmup therefore acts as the discovery phase for private identity subnetworks.

6.1. Limitations and Future Work

Single heterogeneity regime. The current experiments use one controlled non-IID condition: four clients with an 80% dominant class ratio. This is useful for isolating the effect of sparse private identity, but it does not fully characterize robustness across milder or more extreme distributions. A stronger evaluation would repeat the same protocol under multiple skew strengths, such as 60%, 80%, and 90% dominant-class partitions, or under Dirichlet partitions with

several concentration parameters.

Client scale and participation. The notebooks simulate full participation from four clients in every round. This makes the method easy to diagnose, but real FL deployments may include many more clients, partial participation, and dropped rounds. Since FedSLIP keeps private adapters persistent on each client, future work should measure whether sparse identities remain stable when clients participate intermittently.

Backbone and task scope. The present implementation uses DistilBERT on AG News. This is appropriate for a lightweight FL+PEFT benchmark, but it does not establish behavior on larger instruction-tuned language models or generative tasks. Extending FedSLIP to larger backbones would test whether the global-private separation remains effective when attention adapters represent richer language behavior.

Privacy accounting. FedSLIP improves privacy exposure by never transmitting the private adapter or its mask, but the current work does not provide a formal privacy guarantee. Future work should combine the communication design with privacy accounting or differential privacy mechanisms to quantify how much client-specific information remains in the transmitted global LoRA updates.

7. Conclusion

We presented FedSLIP, a federated sparse LoRA method for zero-shot personalization under heterogeneous client data. Starting from a standard FedAvg+LoRA baseline, we identified the collaboration penalty: global averaging preserves shared performance but dilutes client identity. A first dual-adapter improvement showed that separating global and private knowledge is promising, but also showed that naive dual adapters can destabilize the global model and increase communication. The final FedSLIP method resolves these issues with a rank-8 global consensus path, a rank-32 sparse private identity path, separate optimizers, FedProx, gradient detachment, LTH masking, warmup, and strict communi-

Table 6. Measured progression across implementation variants. The First Improvement personalization value is measured after its local personalization phase; FedSLIP personalization is zero-shot.

Variant	Global Acc.	Global-path Client Avg.	Personalized Avg.	Comm./Round
FedAvg+LoRA	90.06	90.43	94.24 after K-step FT	4.50 MB
First Improvement	74.25	74.15	92.28 after K-step FT	11.31 MB
FedSLIP	88.00	88.13	93.89 zero-shot	4.50 MB

cation filtering. On non-IID AG News, FedSLIP achieves 93.89% zero-shot personalized accuracy, nearly matching post-hoc local personalization while preserving the 4.50 MB per-round communication cost of vanilla LoRA. These results support a broader conclusion for federated PEFT: robust personalization requires not only smaller updates, but a principled separation between what is globally shared and what must remain privately local.

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